

0. Definitions

$$h[k] = h_d[k] - \delta h[k], \quad \Delta h_d[k] = h_d[k+1] - h_d[k]$$

$$a'_{hd}[k] := \left. \frac{da_h}{dh} \right|_{h_d[k]}, \quad a_r[k] := a_h[k] - a_h(h_d[k])$$

$$\delta h[k+1] = \lambda_c \delta h[k] - F_{dh}[k] e_{a_h}[k] - \varepsilon_h[k]$$

1. True

$$\begin{aligned} a_h[k+1] &= a_h(h_d[k] + \Delta h_d[k] - \lambda_c \delta h[k] + F_{dh}[k] e_{a_h}[k] + \varepsilon_h[k]) \\ &\approx a_h[k] + a'_{hd}[k] \Delta h_d[k] + (1 - \lambda_c) a'_{hd}[k] \delta h[k] + a'_{hd}[k] F_{dh}[k] e_{a_h}[k] + a'_{hd}[k] \varepsilon_h[k] \end{aligned}$$

$$\boxed{a'_{hd}[k+1] = a'_h(h_d[k] + \Delta h_d[k]) \approx a'_{hd}[k]}$$

$$\boxed{a_r[k] = a_h[k] - a_h(h_d[k]) = -a'_{hd}[k] \delta h[k]}$$

$$a'_{hdm}[k] := +\sqrt{\text{Var}(a_r[k]) / \text{Var}(\delta h[k])} = a'_{hd}[k] \quad (a'_{hd} > 0)$$

$$\boxed{a'_{hd}[k+1] = \beta a'_{hd}[k] + (1 - \beta) a'_{hdm}[k], \quad a'_{hdm} \equiv a'_{hd}}$$

2. $\hat{a}'_{hdm}[k]$

$$\begin{aligned} \bar{a}_h[k+1] &= (1 - a_{var}) \bar{a}_h[k] + a_{var} \hat{a}_h[k+1], \quad \hat{a}_r[k] = \hat{a}_h[k] - \bar{a}_h[k] \\ \hat{\sigma}_{\hat{a}_r}^2[k+1] &= (1 - a_{cov}) \hat{\sigma}_{\hat{a}_r}^2[k] + a_{cov} \hat{a}_r^2[k+1] \\ \overline{\delta \hat{h}}[k+1] &= (1 - a_{var}) \overline{\delta \hat{h}}[k] + a_{var} \delta \hat{h}_3[k+1], \quad \delta \hat{h}_r[k] = \delta \hat{h}_3[k] - \overline{\delta \hat{h}}[k] \\ \hat{\sigma}_{\delta \hat{h}}^2[k+1] &= (1 - a_{cov}) \hat{\sigma}_{\delta \hat{h}}^2[k] + a_{cov} \delta \hat{h}_r^2[k+1] \end{aligned}$$

$$\boxed{\hat{a}'_{hdm}[k] := +\sqrt{\hat{\sigma}_{\hat{a}_r}^2[k] / \hat{\sigma}_{\delta \hat{h}}^2[k]}}$$

3. Self-reference and lumped fidelity κ

$u \equiv 0$ (predict only), $\Delta h_d = 0$:

$$\hat{a}_h[k+1] = \hat{a}_h[k] + (1 - \lambda_c) \hat{a}'_{hd}[k] \delta \hat{h}_3[k], \quad (1 - \lambda_c) \delta \hat{h}_3[j] = \delta \hat{h}_3[j] - \delta \hat{h}_3[j+1]$$

$$\Rightarrow \quad \hat{a}_h[k] = \hat{a}_h[k_0] + \hat{a}'_{hd} (\delta \hat{h}_3[k_0] - \delta \hat{h}_3[k]) \quad \Rightarrow \quad \hat{a}_r[k] = -\hat{a}'_{hd}[k] \delta \hat{h}_r[k]$$

$$\boxed{\hat{a}'_{hdm}[k] \equiv \hat{a}'_{hd}[k] \quad \forall \hat{a}'_{hd}}$$

$u \neq 0$ (u_r = accumulated corrections in $\hat{a}_r$; truth enters u_r only through $\nu_2[k] \approx -e_{a'_h} \delta h_r[k-d] + n_2[k]$):

$$\hat{a}_r[k] = -\hat{a}'_{hd}[k] \delta \hat{h}_r[k] + u_r[k]$$

$$\mathbb{E}[\hat{a}_r^2] = \hat{a}'_{hd}{}^2 \sigma_{\delta h}^2 - 2 \hat{a}'_{hd} \mathbb{E}[\delta \hat{h}_r u_r] + \sigma_u^2, \quad \sigma_u^2 := \mathbb{E}[u_r^2]$$

$$\mathbb{E}[\delta \hat{h}_r u_r] =: -\kappa e_{a'_h} \sigma_{\delta h}^2 \quad (\text{def. of } \kappa)$$

$$\frac{\mathbb{E}[\hat{a}_r^2]}{\sigma_{\delta h}^2} = \hat{a}'_{hd}{}^2 + \underbrace{2 \hat{a}'_{hd} \kappa e_{a'_h}}_{\text{truth}} + \underbrace{\sigma_u^2 / \sigma_{\delta h}^2}_{\text{correction noise}}$$

$$\sqrt{\hat{a}'_{hd}{}^2 + \epsilon} \approx \hat{a}'_{hd} + \frac{\epsilon}{2\hat{a}'_{hd}} :$$

$$\boxed{\hat{a}'_{hdm}[k] = \hat{a}'_{hd}[k] + \kappa[k] e_{a'_h}[k] + n_{var}[k]}$$

$$n_{var}[k] = b_u + w[k]$$

$$b_u := \frac{\sigma_u^2}{2 \hat{a}'_{hd} \sigma_{\delta h}^2} \quad (\text{floor rectification; measured } \hat{a}'_{hdm}{}^2 - \hat{a}'_{hd}{}^2 = 1.02 \times 10^{-7} \Rightarrow b_u \approx +0.02 \hat{a}'_{hd})$$

$$w[k] : \text{finite-window sampling noise of } \hat{\sigma}_{\hat{a}_r}^2, \hat{\sigma}_{\delta \hat{h}}^2, \quad \mathbb{E}[w] = 0, \quad \text{Var}(w) = R_{var}$$

$$R_{var} \text{ (delta method on ratio + square root, with } \text{Var}(\hat{\sigma}^2) = a_{cov} IF(\sigma^2)^2 \text{):}$$

$$\text{relvar}(\hat{a}'_{hdm}) = \frac{1}{4} \left[\text{relvar}(\hat{\sigma}_{\hat{a}_r}^2) + \text{relvar}(\hat{\sigma}_{\delta \hat{h}}^2) - 2 \text{relcov} \right]$$

$$\boxed{R_{var} \approx \frac{a_{cov}}{4} (IF_{\hat{a}_r} + IF_{\delta \hat{h}} - 2 IF_{\times}) \hat{a}'_{hd}{}^2}$$

$$IF_* = 1 + 2 \sum_{j \geq 1} \rho_*(j) (1 - a_{cov})^j, \quad \rho_{x^2}(j) = \rho_x^2(j) \text{ (Gaussian)}, \quad \rho_{\delta h} : \lambda_c \text{ family}$$

$$\hat{a}'_{hdm}[k] = \kappa[k] a'_{hd}[k] + (1 - \kappa[k]) \hat{a}'_{hd}[k] + n_{var}[k]$$

$$\kappa = 1 : \hat{a}'_{hdm} - \hat{a}'_{hd} = e_{a'_h} + n_{var} \quad \kappa \rightarrow 0 : \hat{a}'_{hdm} \equiv \hat{a}'_{hd}$$

Measured (frozen $\hat{a}'_{hd} \in \{0.5, 1, 2\} \times a'_{hd}$, hold $\bar{h} = 2.0$, 8 seeds):

$$\kappa_{emp} = +0.03 \pm 0.04, \quad \hat{a}'_{hdm}{}^2 - \hat{a}'_{hd}{}^2 = 1.02 \times 10^{-7} \text{ arm-constant} \Rightarrow \kappa \approx 0$$

4. Esti

$$\begin{aligned}\hat{a}_h[k+1] &= \hat{a}_h[k] + \hat{a}'_{hd}[k] \Delta h_d[k] + (1 - \lambda_c) \hat{a}'_{hd}[k] \delta \hat{h}[k] \\ \hat{a}'_{hd}[k+1] &= \beta \hat{a}'_{hd}[k] + (1 - \beta) \hat{a}'_{hdm}[k]\end{aligned}$$

5. Error

$$\begin{aligned}e_{a_h} &= a_h - \hat{a}_h, & e_{a'_h} &= a'_{hd} - \hat{a}'_{hd}, & e_h &= \delta h - \delta \hat{h} \\ (a'_{hdm} - \hat{a}'_{hdm} &= (1 - \kappa) e_{a'_h} - n_{var}) : \\ e_{a'_h}[k+1] &= [\beta a'_{hd}[k] + (1 - \beta) a'_{hdm}[k]] - [\beta \hat{a}'_{hd}[k] + (1 - \beta) \hat{a}'_{hdm}[k]] \\ &= \beta e_{a'_h}[k] + (1 - \beta) (a'_{hdm}[k] - \hat{a}'_{hdm}[k]) = [1 - (1 - \beta) \kappa] e_{a'_h}[k] - (1 - \beta) n_{var}[k]\end{aligned}$$

$$\begin{aligned}e_{a_h}[k+1] &= (1 + \hat{a}'_{hd}[k] F_{dh}[k]) e_{a_h}[k] + \Delta h_d[k] e_{a'_h}[k] + (1 - \lambda_c) \hat{a}'_{hd}[k] e_h[k] + \hat{a}'_{hd}[k] \varepsilon_h[k] \\ e_{a'_h}[k+1] &= [1 - (1 - \beta) \kappa[k]] e_{a'_h}[k] - (1 - \beta) n_{var}[k]\end{aligned}$$

6. F_e

$$x_e = [\delta h_1, \delta h_2, \delta h_3, e_{a_h}, e_{a'_h}]^T$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{a}'_{hd}[k] & 1 + \hat{a}'_{hd}[k] F_{dh}[k] & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & 1 - (1 - \beta) \kappa[k] \end{bmatrix}$$

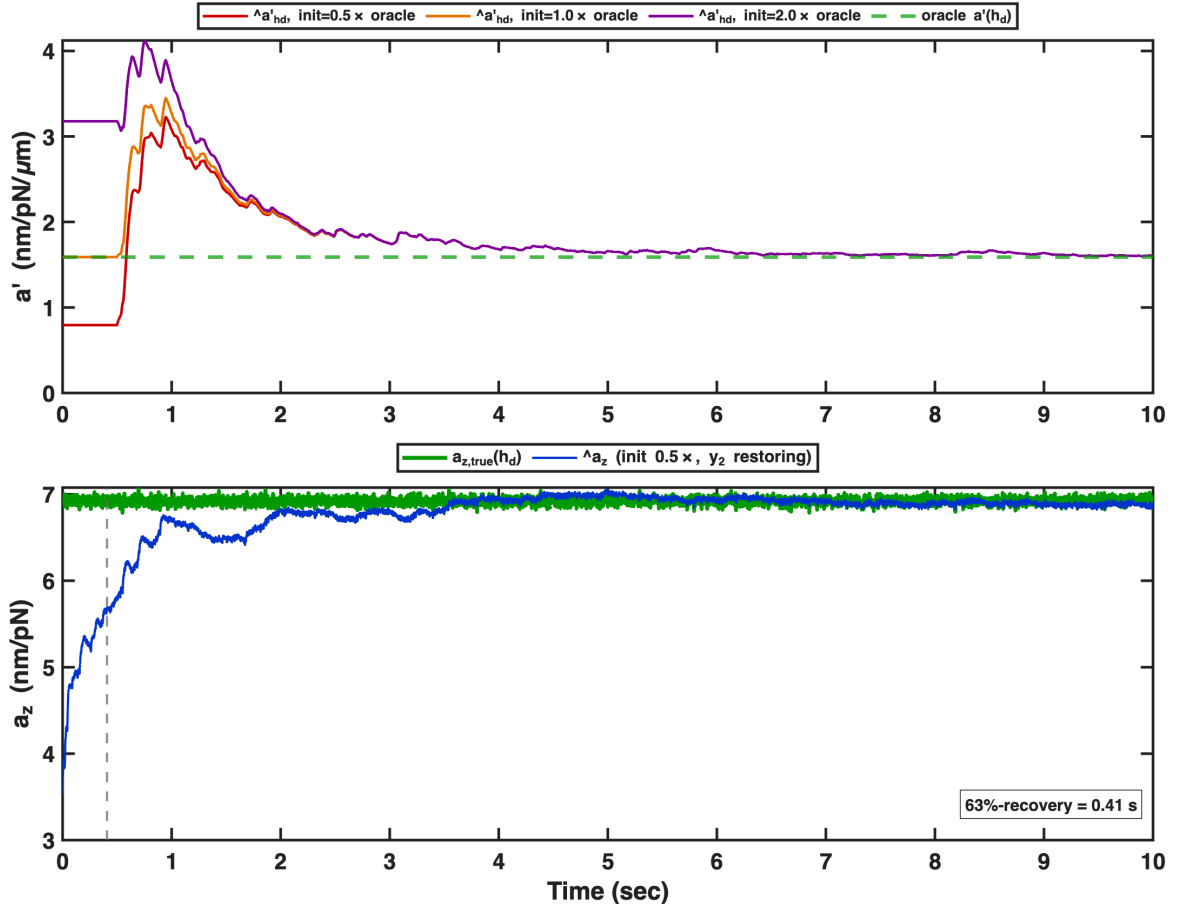
$$q[k] = [0, 0, -\varepsilon_h[k], \hat{a}'_{hd}[k] \varepsilon_h[k], -(1 - \beta) n_{var}[k]]^T$$

7. Q₅₅

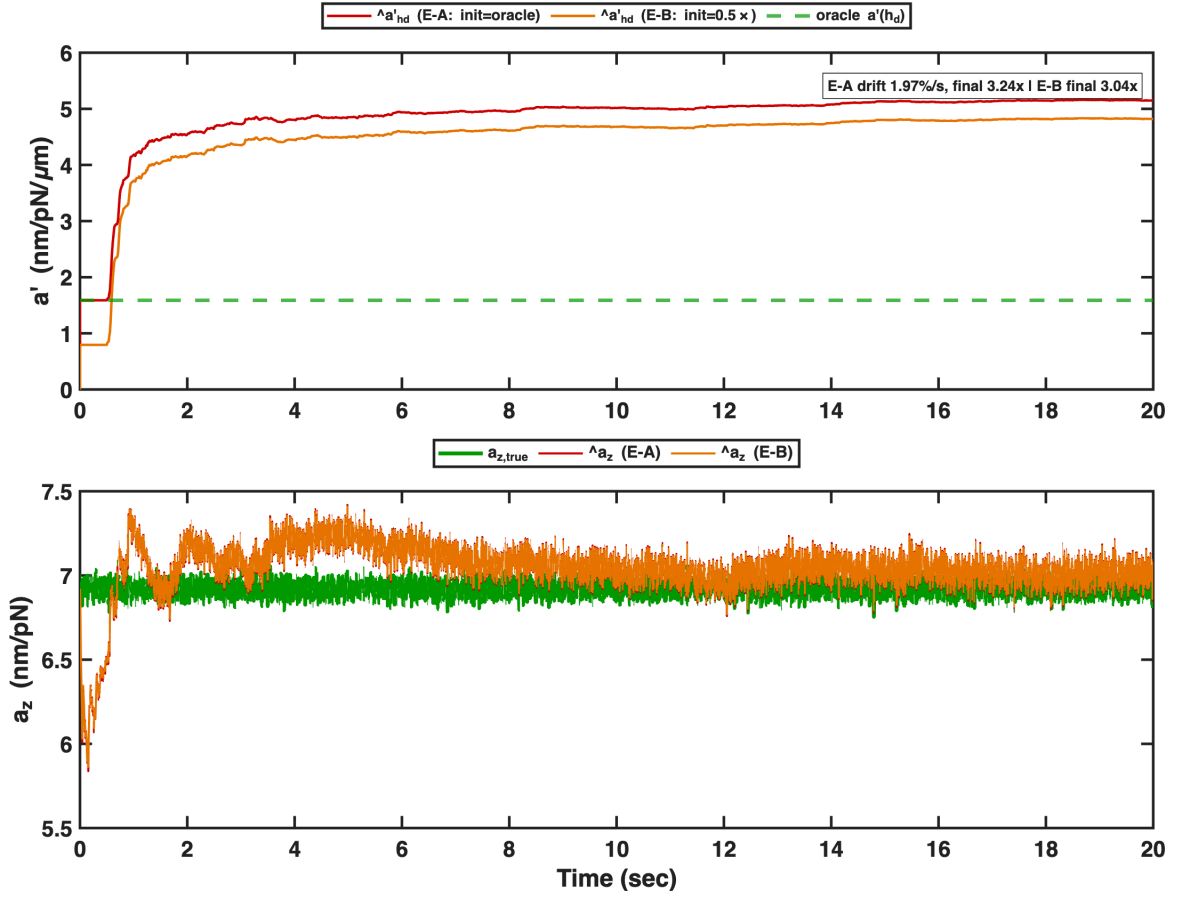
Row-5 process noise (from the $e_{a'_h}$ recursion / $q[k]$):

$$q_5[k] = -(1 - \beta) n_{var}[k] \quad \implies \quad Q_{55} := \text{Var}(q_5) = (1 - \beta)^2 R_{var}$$

8. Simulation



Recovery test (hold $\bar{h} = 2.0$, all corrections on, shadow). Top: wrong-initialized $\hat{a}'_{hd} \in \{0.5, 1, 2\} \times a'_{hd}$ all settle on the signal author's value (the controller's known a') — copying, not measuring (§3; with the author frozen at a wrong value, $\hat{a}'_{hdm}$ follows that value instead, $\kappa_{emp} \approx 0$). Bottom: $\hat{a}_z$ with wrong init recovers through the independent y_2 (genuine restoring force, 63% in 0.41 s).



Closed loop in a hold ($\hat{a}'_{hd}$ fed back, $\Delta h_d = 0$): truth is not a fixed point — both arms (init = a'_{hd} and $0.5\times$) drift to a common wrong attractor $\approx 3 \times a'_{hd}$ (bounded, $+2\%/s$, no descent amplification). $\hat{a}_z$ and tracking stay protected by the independent y_2 ($1.01\times$, 22 nm).